OFFICIAL MODEL SPECIFICATION

# Nexus-70B-Instruct

**Version:** v1.2.0 • **License:** Apache-2.0 with Responsible AI Deployment Addendum • **Type:** Dense Decoder-only Transformer LLM • **Released:** 2026-08-15

Nexus-70B-Instruct is a high-capability generative language model optimized for complex reasoning, multi-turn coding, and instruction following across enterprise workflows.

| BENCHMARKS | PEAK SCORE | MAX DELTA | EVAL RUN ID |
|---|---|---|---|
| **9** | **94.1%** | **+6.7%** | **eval-run-202** |
| Multi-domain | Standardized | vs Nexus-70B-v1 | 2026-08-17 |

## 1. Architecture & Governance

### Model Specifications

**Parameters:** 70 Billion parameters
**Layers:** 80
**Heads:** 64
**Kv Heads:** 8
**Context Window:** 131,072 tokens
**Activation:** SwiGLU
**Positional Embeddings:** RoPE (Rotary Position Embeddings) with theta=500000

### Governance & Ownership

**Organization / Lab:** Nexus AI Research Foundation
**Maintenance Contact:** Nexus Foundation AI Safety & Alignment Group — ai-safety@nexus-models.example.org
**Hardware & Compute:** *Not provided*

## 2. Operational Scope & Limitations

### Primary Intended Applications

• Complex reasoning, quantitative analysis, and mathematical problem solving
• Software engineering assistance and multi-language code generation
• Technical document summarization and executive brief preparation
• Enterprise agentic tool use and workflow orchestration

### Known Limitations

• Context window limited to 128k tokens; attention quality degrades on >100k distractors
• May occasionally hallucinate citations when querying real-time world events beyond knowledge cutoff
• Non-English low-resource language performance is lower than high-resource benchmarks

### Prohibited & Out-of-Scope Deployments

• Autonomous medical diagnosis or clinical prescription without certified human-in-the-loop oversight
• Generation of malicious cyber-attack scripts, malware, or exploit payloads
• High-frequency algorithmic trading decision making without backtesting guardrails
• Automated legal sentencing, credit denial, or biometric surveillance decisions

## 3. Quantitative Evaluation Benchmarks (vs Nexus-70B-v1.1)

Last evaluation cycle: 2026-08-17 10:30:00 UTC • Run ID: eval-run-20260817-v1.2

| Benchmark | Domain | Metric / Setting | Score | Baseline (Nexus-70B-) | Delta | Details |
|---|---|---|---|---|---|---|
| **MMLU (Massive Multitask Language Understanding)** | General Knowledge & Reasoning | Accuracy (5-shot) (5-shot) | **88.4%** | 84.1% | ▲ +4.3% | 57 subjects across STEM, humanities, social sciences |
| **GSM8K (Grade School Math 8K)** | Mathematical Reasoning | Exact Match (Pass@1) (8-shot) | **93.6%** | 89.2% | ▲ +4.4% | Multi-step grade school math word problems |
| **MATH** | Advanced Mathematics | Exact Match (4-shot) | **68.2%** | 61.5% | ▲ +6.7% | Challenging competition math problems |
| **HumanEval** | Code Generation | Pass@1 (0-shot) | **86.6%** | 81.0% | ▲ +5.6% | Python functional code generation evaluated with unit tests |
| **MBPP (Mostly Basic Python Problems)** | Code Generation | Pass@1 (3-shot) | **84.2%** | 80.5% | ▲ +3.7% | Introductory Python programming benchmark |
| **ARC (Challenge)** | Scientific Reasoning | Accuracy (25-shot) (25-shot) | **94.1%** | 91.8% | ▲ +2.3% | Grade-school science questions requiring deep reasoning |
| **HellaSwag** | Commonsense Reasoning | Accuracy (10-shot) (10-shot) | **91.5%** | 89.4% | ▲ +2.1% | Challenging commonsense NLI reasoning |
| **TruthfulQA** | Factuality & Hallucination Resistance | MC2 (0-shot) | **79.8%** | 74.3% | ▲ +5.5% | Factuality & resistance to human misconceptions |
| **AlpacaEval 2.0** | Instruction Following | Win Rate (LC) (0-shot) | **52.4%** | 46.1% | ▲ +6.3% | Length-controlled instruction following benchmark |

## 4. Training Data & Safety Evaluations

### Training Data Summary

**Total Tokens:** 15.2 Trillion tokens
**Composition:** Curated web crawl (55%), Technical papers & textbooks (20%), Synthetic reasoning datasets (15%), High-quality source code (10%)
**Cutoff Date:** March 2026
**Filtering:** Strict toxicity, PII redaction, deduplication, and synthetic quality scoring

### Safety Alignment & Ethics

• Extensively red-teamed against prompt injection, jailbreaking, and hazardous chemical/biological requests
• Reinforcement learning with Human & AI Feedback (RLHF/RLAIF) applied for helpfulness and safety
• Carbon emissions offset verified: 120 tCO2eq neutralized via certified carbon credits

## 5. Citation & Reference

```
@article{nexus2026,
title={Nexus-70B: Technical Report on Frontier Reasoning and Agentic Capabilities},
author={Nexus Research Team},
journal={arXiv preprint arXiv:2608.01234},
year={2026}
}
```